Alex Mercer

Backend Rust Developer

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Professional Summary

Systems-focused Backend Engineer with 4+ years of experience designing and building high-throughput, low-latency distributed systems using **Rust**. Specialized in multi-threaded network programming, asynchronous execution (`tokio`), custom memory allocators, and PyO3/FFI integrations. Proven track record of optimizing microservice memory footprints by up to 60% while maintaining microsecond-level p99 latency SLAs.

Technical Skills

- **Languages:** Rust (Advanced), C/C++, Python, SQL, Shell (Fish/Bash)
- **Rust Ecosystem:** `tokio`, `actix-web`, `axum`, `serde`, `sqlx`, `diesel`, `tonic` (gRPC), `PyO3`, `tracing`
- **Databases & Caching:** PostgreSQL, Redis, ScyllaDB, RocksDB, Vector Databases (Qdrant)
- **Distributed Systems & Messaging:** Apache Kafka, gRPC, Protocol Buffers, RabbitMQ, NATS
- **DevOps & Infrastructure:** Docker, Kubernetes, CI/CD (GitHub Actions), Prometheus, Grafana, Linux (Perf, Valgrind)

Professional Experience

Senior Backend Engineer (Rust) | *Apex Cloud Infrastructure*
San Francisco, CA | **Jan 2024 – Present**

- Architected a high-throughput event processing engine in Rust handling **500k+ events/sec** using `tokio` and lock-free data structures (`crossbeam`), reducing overall CPU utilization by 35%.
- Re-engineered core routing services from Go to Rust, reducing p99 latency from **12ms to 1.8ms** and eliminating garbage collection pause spikes.
- Implemented custom zero-copy deserialization using `serde` and `bytes`, slashing network I/O overhead across internal gRPC microservices.
- Mentored 5 junior engineers in safe memory management, ownership patterns, lifetime annotations, and concurrency primitives (`Arc`, `Mutex`, `rwlock`).

Backend Software Engineer | *Aether Systems*
Remote | **Jun 2022 – Dec 2023**

- Developed production-ready RESTful and gRPC microservices using **Axum** and **Tonic**, communicating with PostgreSQL via `sqlx` compile-time safe queries.
- Built a high-performance Python extension module using **PyO3** to offload CPU-heavy mathematical computations to compiled Rust code, speeding up data pipelines by **8x**.
- Engineered a distributed caching layer utilizing Redis and local `moka` in-memory storage, achieving a **94% cache hit rate** for hot key-value lookups.
- Set up automated CI/CD pipelines incorporating `cargo-clippy`, `cargo-audit`, and property-based testing with `proptest`.

Key Open-Source Projects

`slab-alloc-rs` — *Thread-Safe Custom Arena Allocator*

- *GitHub:* `[github.com/alexmercer-dev/slab-alloc-rs](https://github.com/alexmercer-dev/slab-alloc-rs)`
- Built a high-efficiency memory arena allocator in Rust designed for fixed-size object pools, reducing heap allocation overhead in tight loops.
- Utilized `unsafe` Rust scoped strictly behind safe abstractions, fully verified with `miri` for memory leak and undefined behavior detection.

`hyper-stream-proxy` — *Asynchronous Reverse Proxy*

- *GitHub:* `[github.com/alexmercer-dev/hyper-stream-proxy](https://github.com/alexmercer-dev/hyper-stream-proxy)`
- Developed a zero-copy asynchronous HTTP/2 and gRPC reverse proxy using `hyper` and `tokio`.
- Implemented dynamic load balancing algorithms and full Prometheus metrics collection via `tracing` and OpenTelemetry.

Education

Bachelor of Science in Computer Science

University of California, Berkeley | Graduated 2022

Certifications & Contributions

- **Rust Foundation Contributor:** Submitted bug fixes and documentation improvements to async ecosystem crates (`tokio-util`, `sqlx`).
- **AWS Certified Solutions Architect – Associate** (2023)